Response to the Reviewer Comments

October 2026

Submission ID	bmjph-2026-006196
Article Title:	**Multilevel Determinants of Under-Five Child Undernutrition in Bangladesh: A Nationally Representative Cross-Sectional Study from the Bangladesh Demographic and Health Survey 2022**

Dear Dr. Helen Hilton and Editorial Team, BMJ Public Health,

We sincerely thank the Editorial Office and the three expert reviewers for their valuable feedback and constructive critiques on our manuscript (Manuscript ID: bmjph-2026-006196), entitled "Multilevel Determinants of Under-Five Child Undernutrition in Bangladesh: A Nationally Representative Cross-Sectional Study from the Bangladesh Demographic and Health Survey 2022." We have carefully considered all comments and executed a comprehensive revision to strengthen the manuscript's methodological rigor, analytical precision, and epidemiological interpretation. Detailed point-by-point responses, organized by reviewer and comment, are provided in the tables below.

Editor's Comments	
Comment 1:	**Manuscript title and study design:** Please revise the title of your manuscript to include the study design. This is the preferred format for BMJ Public Health.
Response:	We thank the Editor for this instruction. We have revised the manuscript title to explicitly incorporate the study design in strict compliance with BMJ Public Health formatting guidelines. Revised Title: "Multilevel Determinants of Under-Five Child Undernutrition in Bangladesh: A Nationally Representative Cross-Sectional Study from the Bangladesh Demographic and Health Survey 2022." ***Modifications in Manuscript:*** *Title updated on Page 1 (Lines 1–3), Abstract (Page 1, Lines 12–15), and Manuscript Running Header.*
Comment 2:	**Reporting guidelines checklist:** Please ensure you have indicated the page/line numbers of your manuscript where the relevant information can be found for all items from the STROBE checklist. Do not leave blanks and indicate any items that do not apply to your study design as 'Not Applicable'.
Response:	We have completed the official STROBE Statement checklist for cross-sectional studies in full. Every checklist item has been matched with exact section titles, page numbers, and continuous line numbers (Lines 1–617) from the revised marked manuscript. No items have been left blank, and non-applicable items (such as prospective trial registration or cohort

	follow-up) are clearly designated as 'Not Applicable' with methodological justifications. ***Modifications in Manuscript:*** *Uploaded as a dedicated supplementary document: 'Supplementary File 1_STROBE_Checklist_Cross_Sectional.docx' (with continuous line numbers mapped across all 22 checklist items) and cited in Section 2.1 (Page 5, Lines 156–158; Reference 28).*
Comment 3:	**Sample eligibility criteria:** Please ensure the inclusion and exclusion criteria are clearly stated.
Response:	We have expanded Section 2.1 ('Study Design, Sampling, and Analytical Sample') to detail the complete sample selection cascade. Specifically, we explicitly report that from the 8,784 live births occurring in the five years preceding the survey, 276 deceased children were excluded, and 4,421 children who were absent or not subsampled for anthropometry were excluded. The remaining children were screened for biological plausibility using WHO flags ($\|HAZ\| \leq 6$, $\|WHZ\| \leq 5$, $\|WAZ\| \leq 5$), resulting in a verified, complete-case analytical sample of 4,087 children aged 0–59 months nested within 3,474 households across 670 community clusters. A formal STROBE participant selection flow diagram has also been added as Figure 2. ***Modifications in Manuscript:*** *Section 2.1 (Page 5, Lines 140–164); Figure 2 (Page 6, Lines 165–167).*

Reviewer 1 Comments	
Comment 1:	**Hypothesized determinants across undernutrition categories:** It seems the authors had hypothesized that factors associated with stunting, wasting, and underweight are similar. If this is the case, it should be carefully interpreted with reasons. If this is not the case, choice of assumed factors (e.g., those are included in the models) should be different for each undernutrition categories and should be justified by theories and evidence.
Response:	We thank Reviewer 1 for this essential epidemiological critique. We did not intend to imply that determinants of stunting, wasting, and underweight are uniform. Rather, our conceptual framework (UNICEF, 2021) recognizes that linear growth faltering (stunting), acute ponderal wasting, and composite underweight stem from distinct biological pathways. Our empirical findings strongly substantiate these differences across the three indices: • **Child Age:** A massive driver of chronic stunting (twofold surge at 12–23 months, AOR = 2.38, 95% CI: 1.85–3.06) and progressive underweight (peaking at 48–59 months, AOR = 2.25), but is completely unassociated with acute wasting (p = 0.449). • **Community Clustering:** Highly significant for stunting (ICC = 1.19%,

	MOR = 1.21, p < 0.001) and underweight (ICC = 1.09%, MOR = 1.20, p < 0.001), but negligible for wasting (ICC = 0.00%, MOR = 1.00, p = 0.718), demonstrating that wasting is an acute, episodic household crisis rather than a community-level contextual phenomenon. • **Reproductive Strain / Parity:** Cumulative births (≥4 children ever born) was specifically associated with wasting (AOR = 3.27, 95% CI: 0.91–11.70), reflecting acute resource competition and maternal physiological depletion, whereas it was non-significant for stunting. • **Domestic Infrastructure:** Refrigerator ownership exerted an independent protective association specifically on underweight (AOR = 0.75, 95% CI: 0.62–0.92), underscoring cold-chain food safety. ***Modifications in Manuscript:*** *Section 1 (Introduction, Pages 2–5, Lines 56–138), Section 2.3 (Explanatory Variables, Page 7, Lines 189–220), Sections 3.4–3.6 (Results, Pages 13–14, Lines 315–351; Table 3, Pages 14–17), and Section 4 (Discussion, Pages 17–20, Lines 355–454).*
Comment 2:	**Scientific novelty compared to published BDHS 2022 articles:** There are similar published articles (https://www.nature.com/articles/s41598-025-99288-y, https://link.springer.com/article/10.1186/s12889-026-26793-8) based on same data (BDHS 2022), so the submitted manuscript does not contain high novelty in scientific literature.
Response:	We are grateful to Reviewer 1 for citing these two recently published articles. We have cited both papers and explicitly delineated our distinct conceptual and methodological contributions in both the Introduction and Discussion: **1. Contrast with Tamanna et al. (Scientific Reports, 2025 [25]):** Tamanna et al. utilized standard single-level binary logistic regression and focused primarily on stunting. By completely ignoring the multi-stage cluster sampling design of BDHS, single-level models artificially deflate standard errors and fail to estimate contextual neighborhood effects. In contrast, our study implements a rigorous two-level hierarchical random-intercept modeling framework across 670 Primary Sampling Units (PSUs), decomposing between-cluster variance and proving that over 80% of community-level variation is explained by household socio-environmental conditions. **2. Contrast with Rashid et al. (BMC Public Health, 2026 [26]):** Rashid et al. ('Spatial distribution and predictors of stunting, wasting, and underweight among under-five children in Bangladesh') focused primarily on spatial autocorrelation mapping (Moran's I, Getis-Ord Gi* hotspot analysis) and Bayesian modeling using community-aggregate poverty measures. In contrast, our study provides the definitive frequentist two-level hierarchical model across the complete verified biological sample (N = 4,087), quantifying between-cluster variance via Intraclass Correlation Coefficients (ICC) and Median Odds Ratios (MOR), while uniquely disaggregating domestic food preservation infrastructure (refrigerator ownership) and identifying the critical post-pandemic 12–23 month weaning transition surge.

	3. Simultaneous Evaluation across WHO Standards: Our paper is the first to evaluate stunting, wasting, and underweight simultaneously within a unified sample under identical multilevel specifications, proving that chronic undernutrition is community-clustered whereas acute wasting operates with zero community clustering as an episodic family shock. **[25]** Tamanna T, Mahmud S, Salma N, Hossain MM, Karim MR. Identifying determinants of malnutrition in under-five children in Bangladesh: insights from the BDHS-2022 cross-sectional study. Scientific Reports. 2025;15(1):14336. doi: 10.1038/s41598-025-99288-y. **[26]** Rashid M, Rahman M, Siddiqi NA, Islam R, Miah S, Roy TK, Rahman M. Spatial distribution and predictors of stunting, wasting, and underweight among under-five children in Bangladesh. BMC Public Health. 2026;26(1):1504. doi: 10.1186/s12889-026-26793-8. ***Modifications in Manuscript:*** *Section 1 (Introduction, Pages 4–5, Lines 105–122) and Section 4 (Discussion, Pages 19–20, Lines 431–442).*

Reviewer 2 Comments	
Comment 1:	**Complex survey design and weighting methodology:** Survey design and weighting: The authors mention using sampling weights for the descriptive analysis but do not specify whether the complex survey design (clustering and stratification) was accounted for in the bivariate and multivariate models... Please explain how weights, strata, and primary sampling units were handled.
Response:	We thank Reviewer 2 for this critical methodological observation. In our revised Methods (Section 2.4 and 2.5), we have clearly documented how weights, clusters, and strata were handled: • **Descriptive & Bivariate Statistics:** The normalized survey weight (Weight = V005 / 1,000,000; mean = 1.000, sum = 4,086.93) was applied to all frequency distributions, weighted percentages, and 95% confidence intervals, accounting for unequal selection probabilities across divisions. • **Multilevel Regression Models:** In accordance with established statistical recommendations for complex surveys (Carle, 2009 [21]; Gelman, 2007), clustering was modeled directly via cluster random intercepts (1 | Cluster_ID) across 670 PSUs. Survey stratification variables (administrative division and urban/rural residence) were incorporated directly as fixed-effect covariates in Model III, ensuring unbiased parameter estimation without artificial weight distortion in generalized linear mixed models. **[21]** Carle AC. Fitting multilevel models in complex survey data with design weights: Recommendations. BMC Medical Research Methodology. 2009;9(1):49. doi: 10.1186/1471-2288-9-49. **[Methodological Ref]** Gelman A. Struggles with survey weighting and regression modeling. Statistical Science. 2007;22(2):153–164. ***Modifications in Manuscript:*** *Section 2.4 and Section 2.5 (Pages 7–8, Lines 221–251).*

Comment 2:	**Missing data and sample selection cascade:** Missing data and selection of the final sample: The final analysis looked at 4,087 children out of 8,784 born in the five years before the survey. Details on missing data for each variable are needed... Add a flow chart (STROBE participant flow).
Response:	We have fully detailed the participant selection cascade in Section 2.1 and added Figure 2 (STROBE Participant Flow Diagram): **1. Initial Sample:** 8,784 live-born children in the 5 years preceding the survey. **2. Deceased Children:** 276 children died before interview (B5 = 0) and were excluded. **3. Anthropometry Subsampling & Absence:** In BDHS 2022, anthropometric measurements were performed in a randomized one-third subsample of households; 4,421 children were not measured due to subsampling protocol or temporary absence. 4. Biologically Implausible Flags: Measured children were screened against WHO biological plausibility cutoffs ($\|HAZ\| \leq 6$, $\|WHZ\| \leq 5$, $\|WAZ\| \leq 5$). Zero children exceeded plausible limits. 5. Analytical Sample: 4,087 children aged 0–59 months with 100% complete records across all 56 variables. ***Modifications in Manuscript:*** *Section 2.1 (Page 5, Lines 140–164); Figure 2 (Page 6, Lines 165–167).*
Comment 3:	**Two-level vs. three-level hierarchical structure:** Choice of the multilevel model: The authors thoughtfully applied a three-level model, accounting for clustering at both the community (PSU) and household levels. However, in BDHS data, many households have only one eligible child... discuss whether household-level clustering is adequately supported.
Response:	We are grateful to Reviewer 2 for this perceptive statistical insight. In our analytical sample of 4,087 children, 3,252 out of 3,664 households (88.8%) contributed only a single child (mean: 1.18 children per household). Fitting a three-level model (child nested in household nested in community) resulted in singular fit with near-zero household-level variance (variance < 0.0001). Consequently, we fitted a parsimonious and statistically stable two-level random-intercept logistic model (Level 1: individual child, Level 2: community cluster/PSU across 670 clusters). Likelihood Ratio Tests confirmed highly significant community-level clustering for stunting ($chi2 = 16.36$, $p < 0.001$) and underweight ($chi2 = 10.07$, $p < 0.001$). ***Modifications in Manuscript:*** *Section 2.5 (Page 8, Lines 232–251).*
Comment 4:	**Model selection and parameter interpretation:** Model selection and interpretation: The manuscript discusses AIC, BIC, and log-likelihood across models. In some cases, Model II appears to have a lower AIC than Model III, which would usually make it the preferred model. If the authors choose to focus on Model III, they should explain the rationale.
Response:	We appreciate this astute observation. While Model II possesses a slightly

	lower BIC due to fewer degrees of freedom, Model III was explicitly retained as the definitive analytical model because it provides the most comprehensive and policy-relevant epidemiological evaluation. Model III simultaneously evaluates administrative division disparities and urban-rural gradients while fully controlling for proximal child, maternal, and household determinants, providing actionable targets for resource allocation. ***Modifications in Manuscript:*** *Section 2.5 (Page 8, Lines 232–251); Section 3.3 (Pages 13–14, Lines 301–314).*
Comment 5:	**Removal of causal terminology:** Interpretation of associations: Because this is a cross-sectional study, causal language should be avoided. The manuscript currently uses terms like 'impact,' 'influence,' or 'effects' in several places. These should be replaced with words like 'associated with' or 'odds of.'
Response:	We have conducted a thorough review of the manuscript and eliminated all causal language. Terms such as 'impact,' 'influence,' and 'effect' have been systematically replaced with association-based terminology (e.g., 'associated with,' 'odds of,' 'correlated with,' 'protective association'). ***Modifications in Manuscript:*** *Revised throughout Abstract (Page 1, Lines 7–36), Results (Pages 8–17, Lines 252–354), and Discussion (Pages 17–20, Lines 355–454).*
Comment 6:	**Bivariate chi-square statistics reporting:** Bivariate analysis: Table 2 provides chi-square statistics along with significance symbols, but it would be helpful to also report the degrees of freedom and p-values.
Response:	Table 2 has been updated to report Pearson Chi-square test statistics, degrees of freedom (df), and exact p-values for all 20 explanatory variables across stunting, wasting, and underweight. ***Modifications in Manuscript:*** *Table 2 (Page 12, Lines 298–300); Section 3.2 (Page 12, Lines 285–297).*
Comment 7:	**Potential confounding and UNICEF framework:** Potential confounding: The selection of variables was based on the UNICEF framework, presumably... Discuss unmeasured confounders like dietary diversity and maternal nutrition.
Response:	We have substantially expanded the Discussion (Section 4, Limitations) to address unmeasured potential confounding. We explicitly acknowledge that BDHS 2022 lacked complete data on maternal dietary diversity, infant young child feeding (IYCF) practices, and maternal micronutrient biomarkers across the full under-five sample. ***Modifications in Manuscript:*** *Section 4 (Discussion: Strengths and limitations, Pages 19–20, Lines 443–454).*

Reviewer 3 Comments (Major Revisions)	
Comment 1:	**Inclusion of child age and multicollinearity evaluation:** Covariate

	adjustment: Child age is absent from the models despite its relevance to nutrition outcomes. Incorporating child age into the models is critical. In addition, living children and children ever born are likely to be highly collinear, and birth order may also overlap with these variables. The authors should consider collinearity diagnostics and sensitivity analyses.
Response:	We are immensely grateful to Reviewer 3 for this pivotal critique. We have fully addressed both components: **1. Full Integration of Child Age:** Child age in completed months has been incorporated across all descriptive, bivariate, and multilevel models (categorized into: 0–11, 12–23, 24–35, 36–47, and 48–59 months). This resolved a major omission in earlier drafts. **2. Substantial Model Fit Improvement:** Adding child age improved model fit dramatically (Delta AIC = 49 points for stunting), confirming that chronic stunting surges during the complementary feeding transition (12–23 months: AOR = 2.38; 95% CI: 1.85–3.06). 3. Collinearity Diagnostics & Sensitivity Analysis: Generalized Variance Inflation Factor (GVIF) diagnostics (Table S4) confirmed that multicollinearity is absent across all socioeconomic and environmental assets (all adjusted GVIF^(1/(2*df)) < 1.40; raw GVIF < 2.8). Collinearity was evaluated between children ever born (raw GVIF = 15.11, adjusted GVIF = 3.887) and living children (raw GVIF = 8.98, adjusted GVIF = 2.997). A sensitivity analysis omitting living children confirmed that all model estimates remained robust without altering any conclusions. ***Modifications in Manuscript:*** *Section 2.3 (Page 7, Lines 189–220); Sections 3.4–3.6 (Pages 13–14, Lines 315–351); Table 3 (Pages 14–17, Lines 352–354); Table S4.*
Comment 2:	**Clustering, strata, and survey weights handling:** Survey design: Please explain how weights, strata, and primary sampling units were handled in the descriptive and multilevel models.
Response:	We have expanded Section 2.4 and 2.5: Normalized sampling weights (Weight = V005 / 1,000,000) were incorporated into all descriptive and bivariate calculations. In multilevel logistic models, PSU clustering is directly modeled via community random intercepts (1 | Cluster_ID) across 670 clusters, while survey strata (division and residence) are included as fixed-effect covariates in Model III. ***Modifications in Manuscript:*** *Section 2.4 and 2.5 (Pages 7–8, Lines 221–251).*
Comment 3:	**Multilevel equation and null-model specification:** Multilevel specification: The null-model AIC and log-likelihood values appear to imply that the model is single-level or that the equation needs clarification. Please ensure the equation accurately reflects the random intercept model.
Response:	We thank Reviewer 3 for identifying this. The statistical equation in Section 2.5 has been corrected to explicitly describe the two-level random-intercept model: logit(p_ij) = beta_0 + X_ij * beta + u_j, where u_j ~ N(0, sigma_u^2) across 670 clusters. Single-level null models were also

	estimated to perform formal Likelihood Ratio Tests confirming significant clustering for stunting ($p < 0.001$) and underweight ($p < 0.001$). ***Modifications in Manuscript:*** *Section 2.5 (Page 8, Lines 232–251); Table 3 (Pages 14–17, Lines 352–354); Tables S1–S3.*
Comment 4:	**Sample size justification and selection cascade:** Analytical sample and missing data: Please explain how the final sample of 4,087 children was derived from the total births in the survey.
Response:	We have provided the complete sample selection cascade in Section 2.1 and Figure 2 (STROBE Participant Flow Diagram), detailing the exclusion of deceased children (n = 276) and children not measured due to DHS one-third subsampling (n = 4,421), leaving 4,087 complete cases with verified WHO plausibility. ***Modifications in Manuscript:*** *Section 2.1 (Page 5, Lines 140–164); Figure 2 (Page 6, Lines 165–167).*
Comment 5:	**Checking statistical significance markers:** Statistical reporting and interpretation: Please check all significance markers against the tables. For example, roof material and TV ownership should be reported accurately according to the model results.
Response:	We have verified and re-computed all statistical outputs across Tables 1, 2, and 3. In Model III: roof material (AOR: 1.05, 95% CI: 0.81–1.37) and TV ownership (AOR: 0.98, 95% CI: 0.82–1.18) were not statistically significant after adjustment; all text descriptions in Results and Discussion have been aligned. ***Modifications in Manuscript:*** *Table 2 (Page 12, Lines 298–300); Table 3 (Pages 14–17, Lines 352–354); Section 3.3 (Pages 13–14, Lines 301–314); Section 4 (Pages 17–20, Lines 355–454).*
Comment 6:	**Distinction from published BDHS 2022 multilevel papers:** Contribution and conclusions: Please distinguish the contribution from published BDHS 2022 multilevel analyses, including Rashid et al. (2026). Replace causal interpretations with association-based wording.
Response:	We have distinguished our contribution from Rashid et al. (2026 [26]) and Tamanna et al. (2025 [25]) in the Introduction and Discussion, adopted strictly non-causal language throughout, and aligned our policy implications to proportionate public health strategies. ***Modifications in Manuscript:*** *Section 1 (Introduction, Pages 4–5, Lines 105–122); Section 4 (Discussion, Pages 19–20, Lines 431–442); Section 5 (Conclusion, Page 20, Lines 455–467).*

Reviewer 3 Comments (Minor Revisions)	
Comment 1:	**Subgroup sample sizes and weighted distributions:** Tables and variable definitions: Add subgroup sample sizes, weighted distributions, and prevalence numbers for each category in Tables 1 and 2.
Response:	Table 1 now presents unweighted sample size (n), weighted percentage

	(%), and 95% confidence intervals for every subgroup across stunting, wasting, and underweight. ***Modifications in Manuscript:*** *Table 1 (Pages 9–11, Line 281); Table 2 (Page 12, Lines 298–300).*
Comment 2:	**Wording of reference group comparisons:** Results wording: Match comparisons to the stated reference groups. Maternal age ≥40 years is compared to 15–19, not <40.
Response:	All comparisons have been corrected to strictly reference the designated baseline group (e.g., maternal age ≥40 is compared to 15–19 years). ***Modifications in Manuscript:*** *Sections 3.4–3.6 (Pages 13–14, Lines 315–351).*
Comment 3:	**Standardized terminology and outcome labels:** Terminology and equation: Correct 'wastefulness' to 'wasting,' use outcome-specific labels instead of generic formulas.
Response:	We have corrected all instances of terminology: 'wastefulness' was corrected to 'wasting'; outcomes are distinctly labeled as Stunting (HAZ < -2), Wasting (WHZ < -2), and Underweight (WAZ < -2). ***Modifications in Manuscript:*** *Section 2.2 (Pages 6–7, Lines 178–188) and Section 2.5 (Page 8, Lines 232–251).*
Comment 4:	**Reference list cleanup and STROBE alignment:** References and reporting: Correct mismatched citations, remove duplicate references, clarify figures.
Response:	All citations and references have been thoroughly verified and restructured into a complete 37-reference Vancouver list. The accidental duplicate entry of Saha et al. was removed, and newly required literature (Rashid et al. 2026, Carle 2009, STROBE statement) was integrated with 1-to-1 sequential in-text matching. ***Modifications in Manuscript:*** *References (Pages 21–24, Lines 520–617); Supplementary File 1 (STROBE Checklist).*